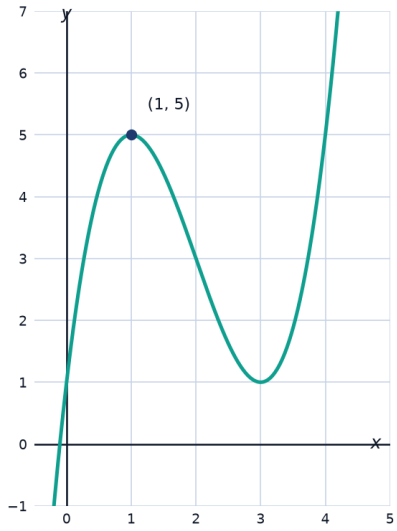
# Differentiation and Rates of Change

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

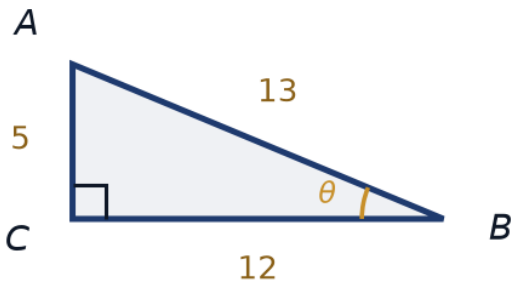
**1** [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

The diagram shows part of the curve $y = ax^3 + bx^2 + 9x + 1$, where $a$ and $b$ are constants. The curve has a stationary point at $(1, 5)$. Find the coordinates of the other stationary point on the curve.




**2** [Foundational] • Target: Product and Quotient Rules

The line $y = 7x - 4$ is tangent to the curve $y = ax^2 + \frac{b}{x}$ at the point where $x = 2$, where $a$ and $b$ are constants and $xe0$. Find the coordinates of the stationary point of the curve and determine its nature.




**3** [Foundational] • Target: Chain Rule and Composite Functions

The variables $x$ and $y$ are connected by the equation $y = \sqrt{2x + 5}$. Given that $x$ increases at a constant rate of $6$ units per second, find the rate of change of $y$ with respect to time when $x = 2$.

**4** [Foundational] • Target: Tangents, Normals and Stationary Points

The variables $x$ and $y$ are connected by the equation $y = 2x^3 - 5x$. Given that $x$ increases at a constant rate of $4$ units $s^{-1}$, find the rate of change of $y$ at the instant when $x = 2$.

© ClipSAT — clipsat.org | All Rights Reserved

**5** [Foundational] • Target: Connected Rates of Change and Applied Optimization

A curve has equation $y = x\sqrt{16 - x^2}$ for $0 \le x \le 4$. A point $P(x, y)$ moves along the curve in the first quadrant, and the normal to the curve at $P$ intersects the $x$-axis at the point $N$. Given that the $x$-coordinate of $P$ is increasing at a constant rate of $\sqrt{3}$ units $s^{-1}$, find the rate of change of the area of triangle $OPN$, where $O$ is the origin, at the instant when $x = 2$.

**6** [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

A storage capsule consists of a right circular cylinder of radius $r$ cm and height $h$ cm, closed with a flat circular base of radius $r$ cm at the bottom, and surmounted by a hemispherical dome of radius $r$ cm at the top. The total exterior surface area of the capsule is fixed at $180\pi$ $cm^2$. During a heating process, the radius $r$ expands at a constant rate of $0.4$ cm $s^{-1}$. Find the rate of change of the total volume of the capsule, in $cm^3 s^{-1}$, at the instant when the height of the cylindrical section is $16.5$ cm.

**7** [Foundational] • Target: Product and Quotient Rules

The variables $x$ and $y$ are related by the equation $y = 2x^3 - 5x$. Given that $x$ is increasing at a constant rate of $3$ units per second, find the rate of change of $y$ with respect to time at the instant when $x = 2$.

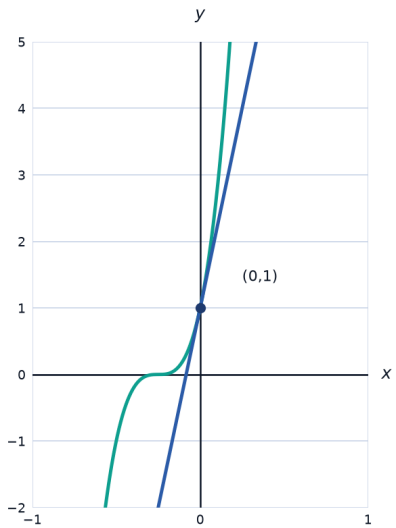
## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Chain Rule and Composite Functions

A curve has equation $y = \frac{kx}{2x - 1}$, where $x \neq \frac{1}{2}$ and $k$ is a non-zero constant. The normal to the curve at the point where $x = 1$ has a gradient of $\frac{1}{3}$. Given that $x$ is increasing at a rate of $0.6$ units per second, find the rate of change of $y$ when $x = 2$.

**9** [Intermediate] • Target: Tangents, Normals and Stationary Points

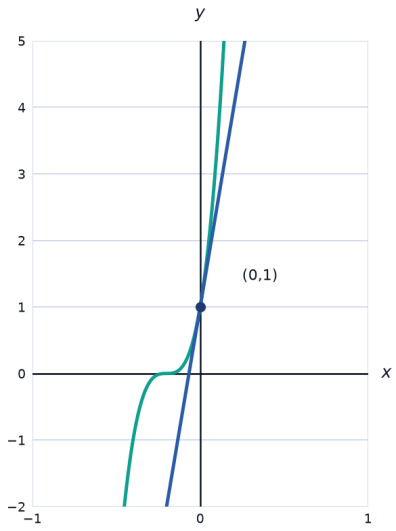
Find the equation of the tangent line to the curve $y = (4x + 1)^3$ at the point where $x = 0$.




© ClipSAT — clipsat.org | All Rights Reserved

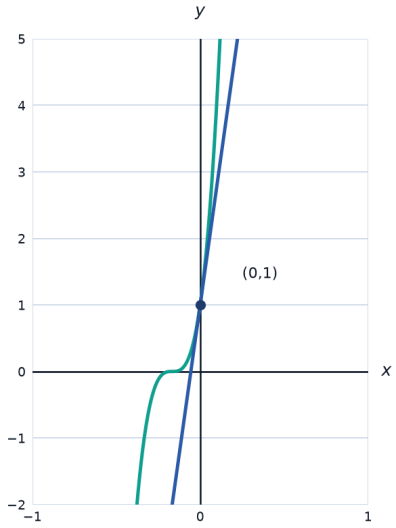
**10** **[Intermediate]** • **Target: Connected Rates of Change and Applied Optimization**

Find the equation of the tangent line to the curve $y = (5x + 1)^3$ at the point where $x = 0$.




**11** **[Intermediate]** • **Target: Differentiation & Rates of Change Practice Exercises**

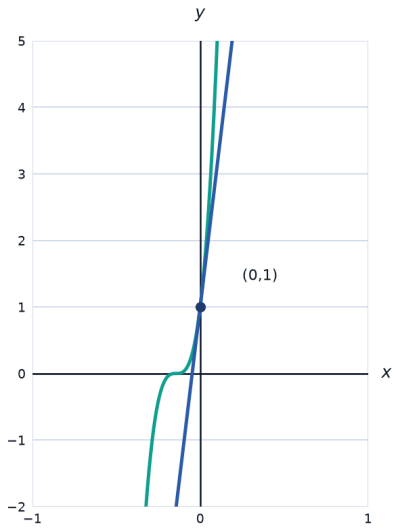
Find the equation of the tangent line to the curve $y = (6x + 1)^3$ at the point where $x = 0$.




© ClipSAT — clipsat.org | All Rights Reserved

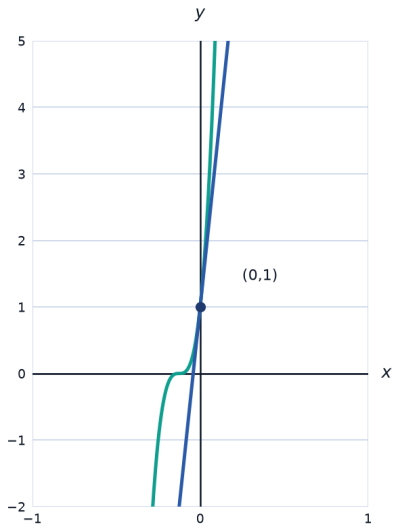
**12** **[Intermediate]** • **Target: Product and Quotient Rules**

Find the equation of the tangent line to the curve $y = (7x + 1)^3$ at the point where $x = 0$.




**13** **[Intermediate]** • **Target: Chain Rule and Composite Functions**

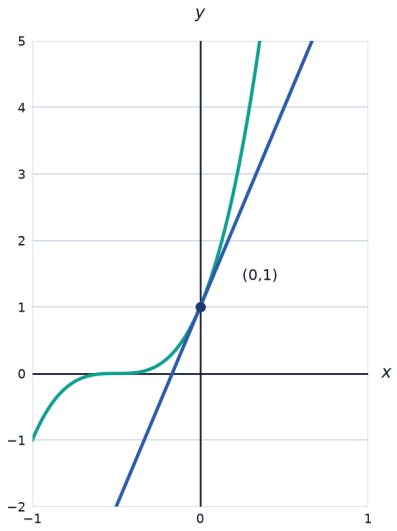
Find the equation of the tangent line to the curve $y = (8x + 1)^3$ at the point where $x = 0$.




© ClipSAT — clipsat.org | All Rights Reserved

**14** [Intermediate] • Target: Tangents, Normals and Stationary Points

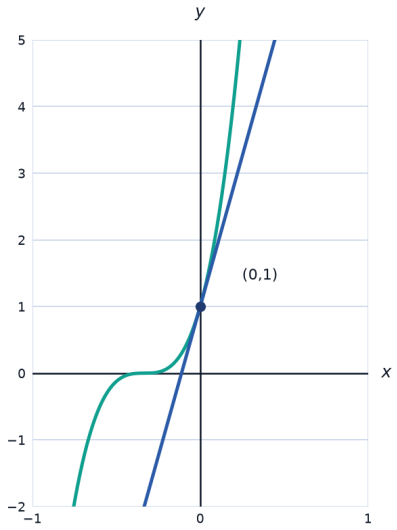
Find the equation of the tangent line to the curve $y = (2x + 1)^3$ at the point where $x = 0$.




## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Connected Rates of Change and Applied Optimization • Cross-Lesson: 8.01 Integration and Kinematics

Find the equation of the tangent line to the curve $y = (3x + 1)^3$ at the point where $x = 0$.




**16** [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

Air is pumped into a spherical balloon such that its radius increases at a constant rate of $0.5$ cm/s. What is the rate of change of volume $\frac{dV}{dt}$ when the radius is $r = 8$ cm? Show your complete algebraic work.

© ClipSAT — clipsat.org | All Rights Reserved

**17** [Challenging] • Target: Product and Quotient Rules • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

Air is pumped into a spherical balloon such that its radius increases at a constant rate of $0.5$ cm/s. What is the rate of change of volume $\frac{dV}{dt}$ when the radius is $r = 9$ cm? Show your complete algebraic work.

**18** [Challenging] • Target: Chain Rule and Composite Functions • Cross-Lesson: 3.01 Indices, Surds and Polynomials

Air is pumped into a spherical balloon such that its radius increases at a constant rate of $0.5$ cm/s. What is the rate of change of volume $\frac{dV}{dt}$ when the radius is $r = 4$ cm? Show your complete algebraic work.

**19** [Challenging] • Target: Tangents, Normals and Stationary Points • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

Air is pumped into a spherical balloon such that its radius increases at a constant rate of $0.5$ cm/s. What is the rate of change of volume $\frac{dV}{dt}$ when the radius is $r = 5$ cm? Show your complete algebraic work.

**20** [Challenging] • Target: Connected Rates of Change and Applied Optimization • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

Air is pumped into a spherical balloon such that its radius increases at a constant rate of $0.5$ cm/s. What is the rate of change of volume $\frac{dV}{dt}$ when the radius is $r = 6$ cm? Show your complete algebraic work.

---

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

Differentiate with respect to \( x \): \( y = 3x^{3} - 5x + 7 \).

**a.** \( \frac{dy}{dx} = 3x^{2} - 5 \)

**b.** \( \frac{dy}{dx} = 9x^{2} \)

**c.** \( \frac{dy}{dx} = 9x^{3} - 5 \)

**d.** \( \frac{dy}{dx} = 9x^{2} - 5 \)

**22** [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

Differentiate with respect to \( x \): \( y = 6x^{4} - 5x + 7 \).

**a.** \( \frac{dy}{dx} = 24x^{3} - 5 \)

**b.** \( \frac{dy}{dx} = 24x^{4} - 5 \)

**c.** \( \frac{dy}{dx} = 6x^{3} - 5 \)

**d.** \( \frac{dy}{dx} = 24x^{3} \)

**23** [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

Differentiate with respect to \( x \): \( y = 9x^{5} - 5x + 7 \).

**a.** \( \frac{dy}{dx} = 45x^{4} - 5 \)

**b.** \( \frac{dy}{dx} = 45x^{5} - 5 \)

**c.** \( \frac{dy}{dx} = 9x^{4} - 5 \)

**d.** \( \frac{dy}{dx} = 45x^{4} \)

© ClipSAT — clipsat.org | All Rights Reserved

**24** [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

Differentiate with respect to \( x \): \( y = 12x^{6} - 5x + 7 \).

**a.** \( \frac{dy}{dx} = 72x^{5} \)

**b.** \( \frac{dy}{dx} = 12x^{5} - 5 \)

**c.** \( \frac{dy}{dx} = 72x^{6} - 5 \)

**d.** \( \frac{dy}{dx} = 72x^{5} - 5 \)

**25** [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

Differentiate with respect to \( x \): \( y = 15x^{7} - 5x + 7 \).

**a.** \( \frac{dy}{dx} = 105x^{6} \)

**b.** \( \frac{dy}{dx} = 15x^{6} - 5 \)

**c.** \( \frac{dy}{dx} = 105x^{7} - 5 \)

**d.** \( \frac{dy}{dx} = 105x^{6} - 5 \)

**26** [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

Differentiate with respect to \( x \): \( y = 18x^{8} - 5x + 7 \).

**a.** \( \frac{dy}{dx} = 144x^{7} \)

**b.** \( \frac{dy}{dx} = 144x^{7} - 5 \)

**c.** \( \frac{dy}{dx} = 144x^{8} - 5 \)

**d.** \( \frac{dy}{dx} = 18x^{7} - 5 \)

**27** [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

Find the equation of the tangent line to the curve \( y = (2x + 1)^3 \) at the point where \( x = 0 \).

**a.** \( y = 2x + 1 \)

**b.** \( y = 6x - 1 \)

**c.** \( y = 6x + 1 \)

**d.** \( y = 6x + 3 \)

**28** [Foundational] • Target: Differentiation & Rates of Change Practice Exercises

Find the equation of the tangent line to the curve \( y = (3x + 1)^3 \) at the point where \( x = 0 \).

**a.** \( y = 9x + 3 \)

**b.** \( y = 9x - 1 \)

**c.** \( y = 3x + 1 \)

**d.** \( y = 9x + 1 \)

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

© ClipSAT — clipsat.org | All Rights Reserved

**29** [Intermediate] • Target: Product and Quotient Rules

Differentiate $y = x^2e^{3x}$ with respect to $x$.

**a.** $xe^{3x}(3x + 2)$

**b.** $2xe^{3x}$

**c.** $3x^2e^{3x}$

**d.** $x^2e^{3x} + 2x$

**30** [Intermediate] • Target: Product and Quotient Rules

Find $\frac{dy}{dx}$ for $y = \frac{2x-1}{x+4}$.

**a.** $\frac{9}{(x+4)^2}$

**b.** $\frac{2}{(x+4)^2}$

**c.** $\frac{7}{(x+4)^2}$

**d.** $\frac{2x+9}{(x+4)^2}$

**31** [Intermediate] • Target: Product and Quotient Rules

Find the derivative of $f(x) = x\ln x$ evaluated at $x = e$.

**a.** 2

**b.** 1

**c.** $e$

**d.** 0

**32** [Intermediate] • Target: Chain Rule and Composite Functions

Differentiate $y = (3x^2 - 5)^4$ with respect to $x$.

**a.** $24x(3x^2 - 5)^3$

**b.** $12x(3x^2 - 5)^3$

**c.** $4(3x^2 - 5)^3$

**d.** $18x^2(3x^2 - 5)^3$

**33** [Intermediate] • Target: Chain Rule and Composite Functions

Find the derivative of $y = \sin^3(2x)$.

**a.** $6\sin^2(2x)\cos(2x)$

**b.** $3\sin^2(2x)\cos(2x)$

**c.** $6\sin(2x)\cos(2x)$

**d.** $-6\sin^2(2x)\cos(2x)$

**34** [Intermediate] • Target: Chain Rule and Composite Functions

Find the gradient of $y = \sqrt{4x + 9}$ at the point where $x = 4$.

**a.** $\frac{2}{5}$

**b.** $\frac{4}{5}$

**c.** $\frac{1}{5}$

**d.** 5

© ClipSAT — clipsat.org | All Rights Reserved

**35** [Intermediate] • Target: Tangents, Normals and Stationary Points

Find the equation of the tangent to the curve $y = x^2 - 4x + 3$ at the point where $x = 3$.

**a.** $y = 2x - 6$

**b.** $y = 2x + 6$

**c.** $y = -2x$

**d.** $y = x - 3$

**36** [Intermediate] • Target: Tangents, Normals and Stationary Points

Find the coordinates of the stationary point of $y = x^3 - 3x + 2$ in the first quadrant.

**a.** $(1, 0)$

**b.** $(1, 2)$

**c.** $(2, 4)$

**d.** $(0, 2)$

**37** [Intermediate] • Target: Tangents, Normals and Stationary Points

Determine the nature of the stationary point of $f(x) = 2x^3 - 9x^2 + 12x$ at $x = 2$.

**a.** Local Minimum

**b.** Local Maximum

**c.** Point of Inflexion

**d.** Vertical Asymptote

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Connected Rates of Change and Applied Optimization • Cross-Lesson: 3.01 Indices, Surds and Polynomials

The radius of a circular oil spill is increasing at $0.5$ m/s. At what rate is the area increasing when the radius is $10$ m?

**a.** $10\pi$ m$^2$/s

**b.** $20\pi$ m$^2$/s

**c.** $5\pi$ m$^2$/s

**d.** $100\pi$ m$^2$/s

**39** [Challenging] • Target: Connected Rates of Change and Applied Optimization • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

A solid cylindrical metal rod of radius $r$ and length $h$ has fixed volume $V = 54\pi$. Express the surface area $S$ in terms of $r$ and find $r$ minimizing $S$.

**a.** $r = 3$

**b.** $r = 2$

**c.** $r = 6$

**d.** $r = 4$

© ClipSAT — clipsat.org | All Rights Reserved

**40** [Challenging] • Target: Connected Rates of Change and Applied Optimization • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

If $y = 3x^2 - 4x$ and $x$ increases at a rate of $2$ units/s, find the rate of increase of $y$ when $x = 5$.

**a.** 52 units/s

**b.** 26 units/s

**c.** 30 units/s

**d.** 60 units/s

**41** [Challenging] • Target: Differentiation & Rates of Change Practice Exercises • Cross-Lesson: 6.01 Permutations and Combinations

Air is pumped into a spherical balloon such that its radius increases at a constant rate of $0.5$ cm/s. What is the rate of change of volume $\frac{dV}{dt}$ when the radius is $r = 7$ cm?

**a.** $98\pi$ cm$^3$/s

**b.** $196\pi$ cm$^3$/s

**c.** $\frac{98}{2}\pi$ cm$^3$/s

**d.** $49\pi$ cm$^3$/s

**42** [Challenging] • Target: Product and Quotient Rules • Cross-Lesson: 8.01 Integration and Kinematics

Air is pumped into a spherical balloon such that its radius increases at a constant rate of $0.5$ cm/s. What is the rate of change of volume $\frac{dV}{dt}$ when the radius is $r = 3$ cm?

**a.** $18\pi$ cm$^3$/s

**b.** $36\pi$ cm$^3$/s

**c.** $9\pi$ cm$^3$/s

**d.** $\frac{18}{2}\pi$ cm$^3$/s

**43** [Challenging] • Target: Chain Rule and Composite Functions • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

Find the $x$-coordinates of the stationary points of $y = x^3 + 15/2x^2 + 18x$.

**a.** $x = -3$ or $x = 2$

**b.** $x = -3$ or $x = -2$

**c.** $x = -2$ or $x = -2$

**d.** $x = 3$ or $x = -2$

© ClipSAT — clipsat.org | All Rights Reserved

**44** [Challenging] • Target: Tangents, Normals and Stationary Points • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

Find the $x$-coordinates of the stationary points of $y = 2x^3 + 12x^2 + 18x$.

**a.** $x = -3$ or $x = -1$

**b.** $x = -2$ or $x = -1$

**c.** $x = 3$ or $x = -1$

**d.** $x = -3$ or $x = 1$

**45** [Challenging] • Target: Connected Rates of Change and Applied Optimization • Cross-Lesson: 3.01 Indices, Surds and Polynomials

Find the $x$-coordinates of the stationary points of $y = 2x^3 - 9x^2 - 24x$.

**a.** $x = 0$ or $x = 4$

**b.** $x = 1$ or $x = 4$

**c.** $x = -1$ or $x = -4$

**d.** $x = -1$ or $x = 4$

© ClipSAT — clipsat.org | All Rights Reserved